

Computer Engineering & Computer Science MPE List (Electives for Specialisation) for AY2025 and later Cohorts

Important Note for CE & CSC MPE:

- Students must take at least 5 major prescribed elective (MPE) courses in each Specialisation area to qualify for the Specialisation as shown in the list below.
- If a student has read at least 5 courses from one Specialisation and a minimum of 15 AUs regardless of whether the courses are read as MPE or Broadening & Deepening Elective (BDE), the student will be deemed to have attained that particular Specialisation. *(Note: Students can only take MPE as BDE after fulfilling the MPE requirements)*
- Courses where the Flexible Grading Option (FGO) has been exercised can be counted towards the award of a Specialisation.
- For courses taken during exchange, up to two (2) Pass/Fail graded MPEs (including equivalent and generic MPEs) may be counted towards a Specialisation. Students may take a maximum of two (2) generic MPEs, of which only one (1) generic MPE may count towards one Specialisation. No double counting of MPEs across Specialisations is permitted.
- Students may be awarded up to two (2) Specialisations. The same MPE course cannot be counted towards more than one Specialisation.
- The "Specialisation" attained will be reflected in the result transcript, e.g., Specialisation in Security.
- Topics to be offered can vary depending on factors such as availability of faculty; availability of visiting staff with certain expertise; new technological trends, etc. Special Topics may also replace the listed specialisation courses. Note that this list is subjected to changes.
- The semester of offering is indicative only. Students should refer to the official [class schedule](#) for the latest course offering information before course registration.
- Courses marked as "**TBC**" indicate that the semester of offering (S1/S2) has yet to be confirmed. Students are encouraged to plan with flexibility and check for updates during the course registration period.

ARTIFICIAL INTELLIGENCE

Course Code	Course Title	AU	Offering		Pre-requisite
			S1	S2	
SC3000	Artificial Intelligence	3	√	√	(SC1007 & SC2000) or (SC1007 & MH2500)
SC4000	Machine Learning	3	√	√	(SC1004 & SC1007 & SC2000) or (SC1007 & MH2500)
SC4001	Neural Networks & Deep Learning	3	√	√	(SC1004 & SC1007) or (MH2802 & SC1007)
SC4002	Natural Language Processing	3	√		SC2001 or MH1403
SC4003	Intelligent Agents	3		√	(SC1007 & SC2000) or (SC1007 & MH2500)
SC4061	Computer Vision	3	√		Year 3 Standing
SC4062	Generative Artificial Intelligence - Advanced Topics	3		√	SC4001 & SC4002
SC4172	Internet of Things: Tiny Machine Learning	3		√	SC2107

SECURITY

Course Code	Course Title	AU	Offering		Pre-requisite
			S1	S2	
SC3010	Computer Security	3	√	√	SC2005
SC4010	Applied Cryptography	3	√		SC2000 or MH2500
SC4011	Security Management	3	√		SC2006
SC4012	Software Security	3		√	SC2002 & SC2005
SC4013	Application Security	3	√		SC2005 & SC2008
SC4014	Concepts & Techniques for Malware Analysis	3	Not offered		SC1006 & SC2005
SC4015	Cyber Physical System Security	3		√	SC1006
SC4016	Cyber Threat Intelligence	3	√		Year 3 Standing
SC4017	Data Privacy & Security	3	√		SC2000 & SC2001
SC4063	Network Security	3		√	SC2005 & SC2008

DATA SCIENCE

Course Code	Course Title	AU	Offering		Pre-requisite
			S1	S2	
SC3020	Database System Principles	3	√	√	SC2207
SC3021	Data Science Fundamentals	3		√	SC1003
SC4000	Machine Learning	3	√	√	(SC1004 & SC1007 & SC2000) or (SC1007 & MH2500)
SC4002	Natural Language Processing	3	√		SC2001 or MH1403
SC4020	Data Analytics & Mining	3	√		SC2001 or MH1403

SC4021	Information Retrieval	3		√	SC2001 or MH1403
SC4022	Network Science	3	Not offered		SC2001 or MH1403
SC4023	Big Data Management	3		√	SC2207
SC4024	Data Visualisation	3	√		(SC1003 & MH2500) or (SC1003 & SC2000)
SC4025	Digital Product Management	3	√		SC2006

HIGH-PERFORMANCE COMPUTING (HPC)

Course Code	Course Title	AU	Offering		Pre-requisite
			S1	S2	
SC3050	Advanced Computer Architecture	3	√		SC1006
SC4023	Big Data Management	3		√	SC2207
SC4050	Parallel Computing	3	√		SC2001
SC4051	Distributed Systems	3		√	SC2005 & SC2008
SC4052	Cloud Computing	3		√	SC1004 or MH2802
SC4054	Simulation & Modelling	3	Not offered		SC1007 & SC2000
SC4055	Introduction to Quantum Computing	3	√		(SC1003, SC1004 & SC2000) or (SC1003, SC2000 & MH2802) or (SC1003, SC1004 & MH2500)
SC4062	Generative Artificial Intelligence - Advanced Topics	3		√	SC4001 & SC4002
SC4064	GPU Programming	3		√	SC2001 & SC2005

EDGE COMPUTING

Course Code	Course Title	AU	Offering		Pre-requisite
			S1	S2	
SC3050	Advanced Computer Architecture	3	√		SC1006
SC3103	Embedded Programming	3	√	√	TBC (Pending Approval)
SC4015	Cyber Physical System Security	3		√	SC1006
SC4030	Wireless & Mobile Networks	3	√		SC2008
SC4031	Internet of Things: Communications & Networking	3		√	SC2008
SC4064	GPU Programming	3		√	SC2001 & SC2005
SC4172	Internet of Things: Tiny Machine Learning	3		√	SC2107

OTHER MPEs

Course Code	Course Title	AU	Offering		Pre-requisite
			S1	S2	
SC3030	Advanced Computer Networks	3	√		SC2008
SC3040	Advanced Software Engineering	3	√		SC2006
SC3060	Computer Graphics & Visualization	3	√		Year 2 Standing
SC3061	Human Computer Interaction	3	√		Year 3 Standing
SC3270	Reasoning About Programs	3	√		SC2001 or SC2301 or (MH1403 & MH1812)
SC3271	Logic, Proof, and the Foundations of Calculus	3	√		SC2203 & MH1812 or SC2203 & SC1124 or SC1124 & MH4302 or MH1812 & MH4302
SC4040	Advanced Topics in Algorithms	3	√		SC2001
SC4053	Blockchain Technology	3	√		SC1007 & SC2001
SC4060	Virtual & Augmented Reality	3	Not offered		Year 3 Standing
SC4242	Compiler Techniques	3	Not offered		SC2001 & SC2006